Decimals and Place Value

Australian Curriculum v9.0 Year 5 Maths
AC9M5N01

Name: ______________________________ Date: __________________

LEARNING TARGET

I can interpret, compare and order decimals beyond hundredths using place value and number lines.

KEY STRATEGY + WORKED EXAMPLE

Compare digits from the greatest place. Add trailing zeros when useful: 4.7 = 4.700. On a number line, locate the decimal between known benchmarks.

Example: 3.407 < 3.47 because 3.407 < 3.470. The first different digit is in the hundredths place: 0 < 7.

A. TRY TOGETHER - USE THE STRATEGY

1. Write 6.305 in expanded form.
2. What is the value of 8 in 14.082?
3. Insert <, > or =: 5.206 ___ 5.26
4. Place 2.375 between two consecutive tenths.

B. CORE PRACTICE - SHOW YOUR THINKING

5. Write 9 + 0.4 + 0.07 + 0.006 as a decimal.
6. Write 12.509 in words.
7. Value of 7 in 3.271?
8. Value of 4 in 18.046?
9. Compare: 7.090 ___ 7.09
10. Compare: 0.805 ___ 0.850
11. Order least to greatest: 4.09, 4.9, 4.099
12. Which is closest to 6.5: 6.49, 6.405 or 6.509?

C. INDEPENDENT PRACTICE - CALCULATE AND CHECK

13. 8.306 ___ 8.360
14. 2.999 ___ 3.001
15. 15.070 ___ 15.07
16. Order: 0.75, 0.705, 0.075
17. Write a decimal between 3.408 and 3.409
18. Round 7.486 to the nearest tenth
19. Round 9.956 to the nearest hundredth
20. Locate 5.625 between consecutive hundredths

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D. ANALYSE & APPLY

21. Three race times are 12.408 s, 12.48 s and 12.084 s. Order fastest to slowest and justify.

22. A rainfall gauge reads 18.375 mm. Explain the value of each digit after the decimal.

23. Correct Noah: '4.62 is less than 4.608 because 62 is less than 608.'

E. REASON, CHECK AND CORRECT

24. Explain why adding zeros to the right of a decimal does not change its value.

25. Which is greater, 0.4 or 0.375? Prove it using place value and a number-line argument.

EXIT TICKET + CHALLENGE

26. Order 6.305, 6.35 and 6.053, then identify the value of the 5 in each number.

Challenge: Create four decimals that all lie between 2.49 and 2.50. Order them and explain how you know.

F. STUDENT REVIEW - SHOW WHAT YOU UNDERSTAND

Use your own words so your teacher can see what you understand and what to practise next.

27. Explain how place value helped you compare two decimals.

28. Verify one comparison using a number line or trailing zeros.

29. Circle one: I understand it | I need more practice | I need help

30. The easiest part was: ____

31. The most difficult part was: ____

Why was it difficult? ____

32. My next practice step is: ____

PARENT / TEACHER TIP

Ask your child to read each decimal aloud and name the value of one digit. Have them justify comparisons from the first differing place.

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TEACHER COPY - ANSWERS AND SUPPORT

ANSWER KEY

A. 1) 6 + 0.3 + 0.005. 2) 0.08 or 8 hundredths. 3) <. 4) Between 2.3 and 2.4.

B. 5) 9.476. 6) twelve and five hundred nine thousandths. 7) 0.07. 8) 0.04. 9) =. 10) <. 11) 4.09, 4.099, 4.9. 12) 6.509.

C. 13) <. 14) <. 15) =. 16) 0.075, 0.705, 0.75. 17) Answers vary, e.g. 3.4085. 18) 7.5. 19) 9.96. 20) Between 5.62 and 5.63.

D. 21) 12.084, 12.408, 12.48. 22) 3 tenths, 7 hundredths, 5 thousandths. 23) 4.620 > 4.608.

E. 24) Trailing zeros do not alter the quantity or place values. 25) 0.400 > 0.375; it lies farther right.

EXIT. 26) 6.053, 6.305, 6.35; the 5 values are 0.05, 0.005 and 0.05 respectively.

Challenge: Answers vary. Check that the student's example follows every condition and that the reasoning is mathematically correct.

F. 27-28) Answers vary. Look for a clear strategy and a valid second check. 29-32) Use the reflection to identify confidence, difficulty and the next teaching step.

COMMON MISCONCEPTIONS

Students may compare decimal digits as whole numbers or assume more digits means a greater value. Align place values and use trailing zeros.

TEACHER CHECK

- [ ] Uses an appropriate Year 5 Maths strategy.
- [ ] Shows or explains thinking clearly.
- [ ] Checks a response using evidence, a rule, or a second method.
- [ ] Identifies a specific difficulty and useful next practice step.

PARENT / TEACHER TIP

Ask the child to explain how they know, not only state an answer. If the explanation is unclear, use small objects, a drawing, or a number line before returning to symbols.

NEXT STEP

Revisit the item the child marked difficult. Model one example, solve one together, then ask the child to complete one independently and explain the strategy.